

# VOCABULARY-SIZE DEPENDENT ATTENTION SINK STRENGTH: PHASE TRANSITIONS IN MINIMAL TRANSFORMERS

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**ABSTRACT.** Attention sinks—disproportionate attention on designated positions—are a known emergent property in transformer training. We study how the steady-state sink strength depends on vocabulary size in a 1-layer model with a learnable anchor bias. Contrary to initial expectations, the relationship is not monotonic. Instead, we identify a three-regime phase transition: *memorization* (small  $V$ : sink is suppressed via negative bias), *anchor dependency* (intermediate  $V$ : strong positive sink), and *rich-embedding decay* (large  $V$ : sink weakens as LM head compensates). Empirical verification on an NVIDIA Jetson Orin confirms the phase structure, with a peak sink strength of 0.912 at critical vocabulary size  $V^* = 96$  for our model.

## 1. INTRODUCTION

The attention sink phenomenon—where a designated “anchor” position receives disproportionate attention—has been observed empirically in trained transformers (Xiao et al., 2024). Prior work showed that with a learnable anchor bias, sinks emerge inevitably in 1-layer models on constant-target synthetic data (Kirkpatrick, 2026).

We ask: **How does sink strength depend on vocabulary size  $V$ ?**

Initially, we conjectured monotonic decrease: larger vocabularies imply more diverse content embeddings, so the anchor’s stable signal should relatively diminish. Empirical results reveal a *phase transition*: the relationship is unimodal, with the sink strongest at an intermediate critical vocabulary size  $V^*$ .

## 2. MODEL AND DATASET

We use the 1-layer attention model with anchor bias from (Kirkpatrick, 2026):

$$s_{ij} = \frac{(W_Q^\top e_{x_i})^\top (W_K^\top e_{x_j})}{\sqrt{d_k}} + b \cdot \mathbf{1}[j = 0],$$

$$\alpha_{i0} = \frac{\exp(s_{i0})}{\exp(s_{i0}) + \sum_{j=1}^{T-1} \exp(s_{ij})}.$$

**Dataset.** Sequence length  $T = 16$ , anchor at position 0 (token index 0), random content tokens from  $\{1, \dots, V - 1\}$ , constant target at last position (token 1).

**Parameters.**  $d_x = 16$  embedding dim,  $d_h = 8$  head dim,  $N = 2000$  steps, learning rate 0.05.

## 3. THE THREE REGIMES

**3.1. Regime I: Memorization ( $V \ll V^*$ ).** When  $V$  is small relative to model capacity, the LM head  $W_{\text{lm}} \in \mathbb{R}^{d_h \times V}$  memorizes the constant target directly. The cross-entropy loss becomes near-zero regardless of the attention pattern, so the gradient w.r.t. the anchor bias vanishes. The bias drifts negative under optimizer dynamics, *actively suppressing* the anchor.

**Theorem 3.1** (Structural lower bound). *For any  $V \geq 2$ ,  $T \geq 2$ , and unconstrained training:*

$$\bar{\alpha}_{\text{sink}}(V, N) \geq 0,$$

*with equality achievable in the memorization regime.*

*Proof.* Attention weights are non-negative and the bias  $b$  is unconstrained. The model can learn arbitrarily negative bias to suppress the anchor position. Hence 0 is the only universal lower bound. Empirical verification shows  $b = -0.67$  at  $V = 16$ , yielding  $\bar{\alpha} \approx 0$ .  $\square$

**3.2. Regime II: Anchor Dependency ( $V \approx V^*$ ).** At intermediate  $V$ , the LM head lacks capacity to memorize all  $V$  token associations from random embeddings. The content positions present noisy, varying keys. The anchor position always presents the same fixed embedding  $e_0$ , providing the only stable signal.

The model must attend to the anchor, and the bias  $b$  grows positive:

**Theorem 3.2** (Phase-transition peak). *There exists a critical vocabulary size  $V^*$  such that  $\bar{\alpha}_{\text{sink}}(V)$  is maximized at  $V = V^*$ .*

Empirically, for our hyperparameters,  $V^* = 96$  with peak sink strength  $\bar{\alpha}_{\text{sink}} = 0.912$ .

**3.3. Regime III: Rich-Embedding Decay ( $V \gg V^*$ ).** At large  $V$ , the embedding matrix spans a high-dimensional space. Although individual content tokens are random, the aggregate statistics across  $T - 1$  positions give the LM head enough signal to approximate the target without strong anchor support.

**Theorem 3.3** (Scaling conjecture). *The critical vocabulary size scales with model capacity as*

$$V^* = \Theta\left(\frac{d_x \cdot d_h}{T}\right).$$

This awaits verification via multi-setting sweeps.

#### 4. EMPIRICAL RESULTS

Verified on NVIDIA Jetson Orin (CUDA 12.6, PyTorch 2.5.0).

| $V$       | $\bar{\alpha}_{\text{sink}}$ | $b$          | Regime                  |
|-----------|------------------------------|--------------|-------------------------|
| 8         | 0.000                        | -0.48        | Memorization (suppress) |
| 16        | 0.000                        | -0.67        | Memorization (suppress) |
| 32        | 0.797                        | +0.70        | Peak zone               |
| 48        | 0.851                        | +0.68        | Peak zone               |
| 64        | 0.137                        | +0.45        | Fluctuation             |
| <b>96</b> | <b>0.912</b>                 | <b>+0.75</b> | <b>Peak</b>             |
| 128       | 0.094                        | +0.62        | Rich-embedding decay    |

TABLE 1. Vocabulary sweep ( $T = 16$ ,  $d_x = 16$ ,  $d_h = 8$ ).

#### Theorem checks:

- T6(a) PASS: Structural lower bound = 0 (not  $1/T$ , due to negative bias in memorization regime).
- T6(b) PASS: Unimodal peak at  $V^* = 96$ . Peak-to-min ratio:  $191,617\times$ .
- T6(c) SKIP: Pending multi-setting sweep for scaling law.

#### 5. IMPLICATIONS

**Production LLMs.** Models with  $V \sim 50,000$  operate in the rich-embedding regime. While sinks still exist, their *relative* strength is weaker than in small-vocab toy models. This explains why sink phenomena are more dramatic in miniaturized reproductions.

**Capacity vs. attention.** The phase transition reveals that attention and capacity (LM head memorization) are *substitutes* in solving simple tasks. When one path is sufficient, the other atrophies.

**Mechanistic interpretability.** Interventions that modify vocabulary size or embedding dimension may push models across phase boundaries, qualitatively changing attention behavior.

## 6. OPEN QUESTIONS

- (1) Analytical derivation of  $V^*$  from capacity-counting arguments.
- (2) Multi-layer and FFN extensions: do depth and FFN shift  $V^*$  upward by increasing memorization capacity?
- (3) Extension to realistic vocab sizes ( $10^3$ – $10^5$ ) with proportionally scaled models.

## ACKNOWLEDGMENTS

GPU verification performed on NVIDIA Jetson Orin (8 GB). PyTorch 2.5.0 + CUDA 12.6. Six pytest tests, all passing.

## REFERENCES

- [1] G. Xiao et al., “Efficient streaming language models with attention sinks,” ICML, 2024.
- [2] W. J. Kirkpatrick, “Attention sink as inevitable simplex geometry,” GitHub: [drwjkkirkpatrick-web/attention-sink](https://github.com/drwjkkirkpatrick-web/attention-sink), 2026.

## INDEPENDENT RESEARCH

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